

Interpretable and Fair Machine Learning for Health-Cost Prediction and Risk Adjustment: A Reproducible Benchmark on Public Data and an Open-Source Toolkit

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Abstract

Background. Risk-adjustment formulas set what health plans are paid for each enrollee. A formula that underpredicts a group's spending gives plans a reason to avoid it, and one that responds strongly to recorded diagnoses rewards intensive coding. Machine-learning methods promise accuracy; their fairness and coding sensitivity are rarely evaluated together on public data.

Methods. We built a prospective benchmark from five two-year MEPS panels (43,568 persons, 2018 to 2023): year-1 demographics, diagnoses and use predict year-2 total spending. Nine models, from weighted least squares to gradient boosting and a combined actuarial neural network, were evaluated by cluster-aware cross-validation on accuracy, group net compensation, an accuracy-fairness frontier, SHAP drivers and response to simulated coding intensity. The code is released as a Python package.

Results. Gradient boosting reached an out-of-sample R^2 of 0.195 (95% interval 0.161 to 0.245) against 0.125 for weighted least squares. Every model underpaid people who need help with daily activities, by \$4,949 to \$9,508 per person-year on a mean cost of \$29,881. A stacked estimator feeding a cross-fitted boosting score into a constrained regression cut that gap to \$210 at an R^2 of 0.178; constraining the linear formula cost a quarter of its R^2 . Both constraints shifted \$2,156 to \$3,342 of overpayment onto enrollees aged 65 and over. Boosting raised its prediction by \$693 per added chronic code; the linear formula by \$1,292.

Conclusions. Flexible models are more accurate, no less fair, and less coding-sensitive than the linear standard here. Fairness constraints work but move money between groups and must be reported for all groups.

Keywords. risk adjustment; machine learning; algorithmic fairness; net compensation; coding intensity; SHAP; MEPS

1. Introduction

Risk adjustment moves several hundred billion dollars a year. In Medicare Advantage, plan payments are scaled by a risk score built from enrollee demographics and diagnoses; in the Affordable Care Act individual and small-group markets, risk scores drive transfers between insurers (Pope et al., 2004; Kautter et al., 2014). The formulas behind these scores are linear regressions of spending on demographic cells and hierarchical condition categories, and their accuracy, measured as R^2 on individual spending, sits between 0.1 and 0.3.

Three problems with these formulas are well documented and are usually studied separately. First, they leave predictable variation on the table, and flexible machine-learning methods can recover some of it (Rose, 2016; Duncan, Loginov and Ludkovski, 2016; Irvin et al., 2020; Andriola et al., 2024). Second, they underpredict spending for identifiable groups, which gives plans a financial reason to avoid those groups; constrained and penalized regressions can remove the underprediction at a cost in overall fit (Zink and Rose, 2020; McGuire,

Zink and Rose, 2021; Bergquist et al., 2019). Third, they respond to what is recorded rather than what is true, so more thorough diagnosis coding raises payment without raising cost. The Medicare Payment Advisory Commission has repeatedly attributed a large part of the gap between Medicare Advantage payments and fee-for-service spending to coding intensity (MedPAC, 2025), and the phase-in of the V28 CMS-HCC model removed several discretionary condition categories in response.

This paper evaluates the three properties together, on public data, with released code. Using five two-year panels of the Medical Expenditure Panel Survey (MEPS), we predict each person's next-year spending from current-year information and compare nine models: weighted least squares in the form payment formulas take, a Tweedie generalized linear model, a two-part model, an elastic net, gradient boosting with two objectives, a random forest, a feed-forward network and a combined actuarial neural network. We ask four questions.

- RQ1. How much more accurately do flexible models predict next-year spending than the linear payment form, and where in the distribution do the gains come from?
- RQ2. Which groups does each model under- or overpay, and how much accuracy does it cost to close the gaps?
- RQ3. What drives the flexible models' predictions, and are the drivers stable and sensible?
- RQ4. How much does each model's payment rise when diagnosis coding becomes more intensive?

Two design choices distinguish the benchmark from most prior work. Cross-validation keeps MEPS primary sampling units whole, so no neighbourhood contributes to both training and test data; ignoring this overstates out-of-sample accuracy for any model that can memorize local effects. And race and ethnicity never enter a model: they are used only to evaluate what a model does to each group.

The main results are that gradient boosting improves R^2 from 0.125 to 0.195 with the gains concentrated in calibration of the top spending decile; that every model, flexible or not, underpays people who need help with activities of daily living by several thousand dollars a year; that a stacked estimator which feeds a cross-fitted boosting score into a constrained regression closes those gaps at a small accuracy cost, while constraining the linear formula directly costs a quarter of its R^2 ; that closing a gap for one group opens one for another; and that the linear formula, not the flexible models, is the most responsive to added diagnosis codes. The last point runs against a common assumption and follows from how each model uses the same information.

2. Background

Payment formulas. The CMS-HCC and HHS-HCC models regress annual spending on age-sex cells and a set of hierarchical condition categories derived from diagnosis codes, with weighted least squares and coefficients constrained to be non-negative (Pope et al., 2004; Kautter et al., 2014). The linear additive form is chosen for transparency and for ease of administration. Ellis, Martins and Rose (2018) review the design choices and their evaluation.

Machine learning for cost prediction. Rose (2016) proposed an ensemble framework for plan payment and found modest gains over the linear standard on Truven claims. Duncan, Loginov and Ludkovski (2016) compared regression frameworks including boosting for health-cost prediction. Irvin et al. (2020) added social determinants to machine-learning risk adjustment. Andriola et al. (2024) introduced a learning algorithm for payment formulas that respects the constraints of a payment system. Wüthrich and Merz (2023) give the actuarial foundations of the models used here, and Schelldorfer and Wüthrich (2019) the combined actuarial neural network, in which a GLM's linear predictor is a fixed skip connection and a network learns a correction.

Fairness. Zink and Rose (2020) defined net compensation, the difference between what a formula pays for a group and what the group costs, and proposed penalized regressions that shrink it. McGuire, Zink and Rose (2021) compared constrained regression, reinsurance and variable selection. Obermeyer et al. (2019) showed

that a widely used algorithm trained on cost as a proxy for need systematically underrated Black patients, which is the reason the outcome here is spending and the interpretation is payment, not need.

Coding intensity. MedPAC (2025) estimated Medicare Advantage risk scores well above what the same beneficiaries would carry in fee-for-service, and CMS's V28 model removed condition categories judged discretionary. The sensitivity of a payment formula to added codes is a property of the formula that can be measured directly, and Section 6.4 does so.

3. Data

Panels. MEPS follows each panel of households for two calendar years. We used the two-year longitudinal files for panels 23 to 27, covering 2018-2019 through 2022-2023, and the Medical Conditions file for each panel's first year. Panels 23 and 24 were later extended to three and four years; we used their original two-year files so that every person contributes one prospective pair with the two-year longitudinal weight.

Sample. A person enters if present in both years with a positive longitudinal weight and alive at the end of year 2 (Table A1). People who died in year 2 have partial-year outcomes; they are excluded from the primary sample and returned in robustness. The analysis sample is 43,568 persons representing 324.4 million a year.

Outcome. Year-2 total expenditure from all payers, in 2024 dollars using the CPI-U. The weighted mean is \$7,538, the median \$1,486, 15.0% spent nothing, and the top 5% of spenders account for 50.0% of spending (Table 1).

Predictors. All from year 1. Demographics: age band by sex and Census region. Conditions: AHRQ Clinical Classifications Software Refined (CCSR) categories from the conditions file, as binary flags for the 124 categories held by at least 0.5% of the sample (Table A6), plus counts of categories and body systems. MEPS assigns body-system placeholder codes (XXX000) where a diagnosis cannot be placed in a category; these are kept in the body-system count and excluded from flags and condition counts. Use: office visits, emergency visits, inpatient discharges, prescription fills and total spending, logged. Perceived health and mental health are taken from round 2 and help with activities of daily living (ADL) and instrumental activities (IADL) from round 1, because round 3 is interviewed in year 2 and would leak year-2 information.

Four nested feature sets follow the spec: F1 demographics; F2 adds conditions; F3 adds prior use and spending; F4 adds income, insurance, perceived health and functional help. F3 is the primary set. Payment formulas in practice exclude prior spending, so F2 is the set closest to a deployable formula and is reported throughout.

Groups for evaluation. Income below 200% of the federal poverty level; Hispanic, non-Hispanic Black and non-Hispanic Asian; needs help with ADL or IADL; any mental or behavioural condition (CCSR MBD); age 65 and over; uninsured all year. Race and ethnicity are never model inputs. Three groups are targets for the fairness estimators: low income, mental health conditions and functional help, chosen because payment literature identifies them as under-compensated and because they are observable to a payer.

CCSR limitation. MEPS releases three-digit ICD-10-CM codes. CCSR categories built from them are coarser than the hierarchical condition categories of the CMS and HHS models, so the linear benchmark here approximates the form of a payment formula, not any specific formula.

4. Methods

4.1 Models

All models share one interface and are fitted with the MEPS longitudinal weight. WLS is weighted least squares on the features, the additive form of a payment formula. The Tweedie GLM has a log link and variance power 1.5. The two-part model multiplies a logistic probability of any spending by a gamma GLM for the

amount. The elastic net is fitted on raw spending. Gradient boosting uses LightGBM with a Tweedie objective and, separately, a squared-error objective, the objective payment formulas optimize. The random forest uses 300 trees. The neural network has two hidden layers, a log link and Tweedie deviance loss. The combined actuarial neural network (CANN) adds the fitted Tweedie GLM's linear predictor as a fixed offset and initializes the network's output layer at zero, so it starts at the GLM and moves only where validation deviance improves.

Multiplicative models do not satisfy the balance property that payments equal costs in total. Each is rescaled on its own training data so that weighted predictions sum to weighted spending, the correction any payment system applies.

Every hyperparameter is chosen inside the training data of each outer fold. The GLM, two-part and elastic-net penalties are chosen by three inner cluster-disjoint folds; boosting and the networks use one inner split of whole clusters for early stopping and a small grid. No test observation influences any choice (Table A5 reports what was chosen).

4.2 Survey-aware cross-validation

Folds are formed from primary sampling units within strata within panel: PSUs are shuffled and dealt across five folds, so every fold draws from every stratum and no PSU is in both a training and a test set. The procedure is repeated three times with different assignments. Each person receives one out-of-fold prediction per repeat, and metrics are computed on the repeat-averaged predictions with weights. Intervals come from a bootstrap that resamples PSUs within strata, 200 replicates; the between-repeat spread of R^2 is reported alongside and is small (at most 0.008).

4.3 Metrics

Accuracy: weighted R^2 , mean absolute error, Cumming's prediction measure (the absolute-error analogue of R^2), predictive ratios by decile of predicted spending, and the share of the true top 10% of spenders placed in the predicted top 10%.

Fairness, following Zink and Rose (2020): for group G , the predictive ratio PR_G is weighted predicted over weighted observed spending, and net compensation NC_G is the weighted mean of prediction minus outcome. Negative NC_G means the group is underpaid.

4.4 Fairness estimators

Because NC_G is linear in the coefficients of a linear model, two estimators have exact solutions. Penalized WLS adds $\lambda \sum_G NC_G^2$ to the weighted squared-error loss, with λ scaled by the outcome variance so it is unit-free; constrained WLS imposes $NC_G = 0$ for each target group, the $\lambda \rightarrow \infty$ limit, solved as a KKT system. Both are fitted over a grid of λ within the survey folds.

For boosting we use a stacked form: a LightGBM score is cross-fitted within the training data (three cluster-disjoint folds, so no training row's score was fitted to its own outcome), and the score enters a penalized or constrained regression together with the features. We tried adding the net-compensation penalty inside the boosting objective and rejected it: the penalty's curvature is shared across a group, a diagonal Hessian either understates it (Newton steps diverge) or, if spread across members, damps the fitting of their own residuals, and in testing under-compensation of small high-cost groups worsened as the penalty rose. The stacked form keeps the boosted model's accuracy in the score and puts the adjustment where it has an exact solution.

As a group-neutral post-processing comparison, the Tweedie LightGBM's predictions are recalibrated by decile of predicted spending, with factors estimated on the other folds' out-of-fold predictions.

Group membership enters estimation only through group means. It never enters the prediction formula, so a fair model pays by the same features as an unconstrained one.

4.5 Drivers

TreeSHAP values are computed for the squared-error LightGBM on up to 4,000 held-out persons per fold, from the model fitted on that fold's training PSUs; the squared-error model is explained because its SHAP values are in dollars, and its accuracy equals the Tweedie model's. Global importance is the weighted mean absolute SHAP value; stability is the Spearman correlation of importance ranks across folds. For the CANN, permutation importance on the held-out fold is used. Two sense checks test whether SHAP contributions rise with year-1 spending and with the number of conditions.

4.6 Coding sensitivity

For a share of held-out persons (5%, 10% or 25%), one or two condition flags they did not have are switched on, drawn either in proportion to each category's prevalence among people with any condition or from a chronic, payment-relevant pool (diabetes, coronary disease, dysrhythmia, depression, COPD and others listed in the code; five of the twelve are common enough to be features). The condition and body-system counts are updated; use and spending are unchanged. Models fitted on each fold's training PSUs are applied before and after, and we report the rise in total predicted spending and the dollar rise per added code.

4.7 Robustness

Each variant changes one decision and reruns WLS, the Tweedie GLM and Tweedie LightGBM on one repeat: outcome capped at \$250,000; a log-outcome linear model; unweighted training; condition-flag prevalence floors of 0.2% and 1%; feature sets F2 and F4; decedents included; panels whose outcome year fell in 2020 or 2021 dropped; and the subsets aged 65 and over and under-65 privately insured.

5. Software

The estimators, folds, metrics, fairness methods and coding stress test are released as `riskfair`, a Python package with no dependence on this paper's configuration: every choice is an argument, so it runs on any person-level cost data with weights and design identifiers, survey or claims. The pipeline that produces every table and figure here calls the package. Unit tests cover fold disjointness, metric identities, exact zero net compensation for the constrained estimator, monotone response to the penalty, recovery of the constraint by the stacked estimator on simulated data, and the coding perturbation. A test asserts that no feature name refers to race or to the year-2 outcome and that no feature is nearly collinear with the outcome.

6. Results

6.1 Accuracy (RQ1)

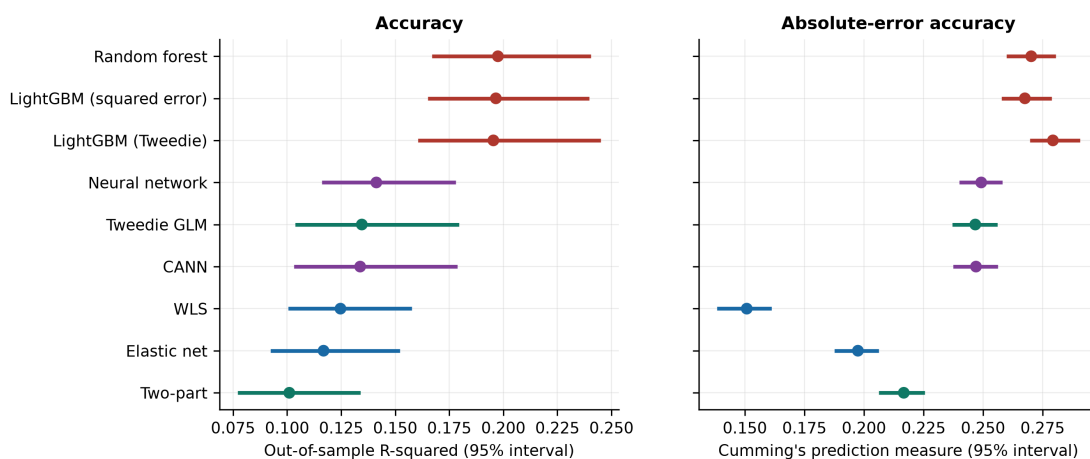


Figure 1. Out-of-sample R^2 and Cumming's prediction measure by model, primary feature set, with 95% intervals from a bootstrap over primary sampling units.

Table 2 and Figure 1 give out-of-sample accuracy on the primary feature set. The three tree ensembles reach R^2 of 0.195 to 0.197 with intervals of roughly 0.16 to 0.25. WLS reaches 0.125 (0.101 to 0.158), the Tweedie GLM 0.135, the two-part model 0.101 and the elastic net 0.117. The neural network reaches 0.141 and the CANN 0.134, indistinguishable from its GLM base: given the same features, the network correction found little to add. Cumming's prediction measure orders the models the same way, with the GLM family closer to the trees on absolute error than on squared error.

The gain from boosting is concentrated where payment formulas are weakest. The predictive ratio in the top decile of predicted spending is 0.93 for WLS and 0.98 for Tweedie LightGBM (Figure 2 and Table A2), and the top-10% capture rate rises from 44% to 48%. Mean absolute error falls from \$7,978 to \$6,770.

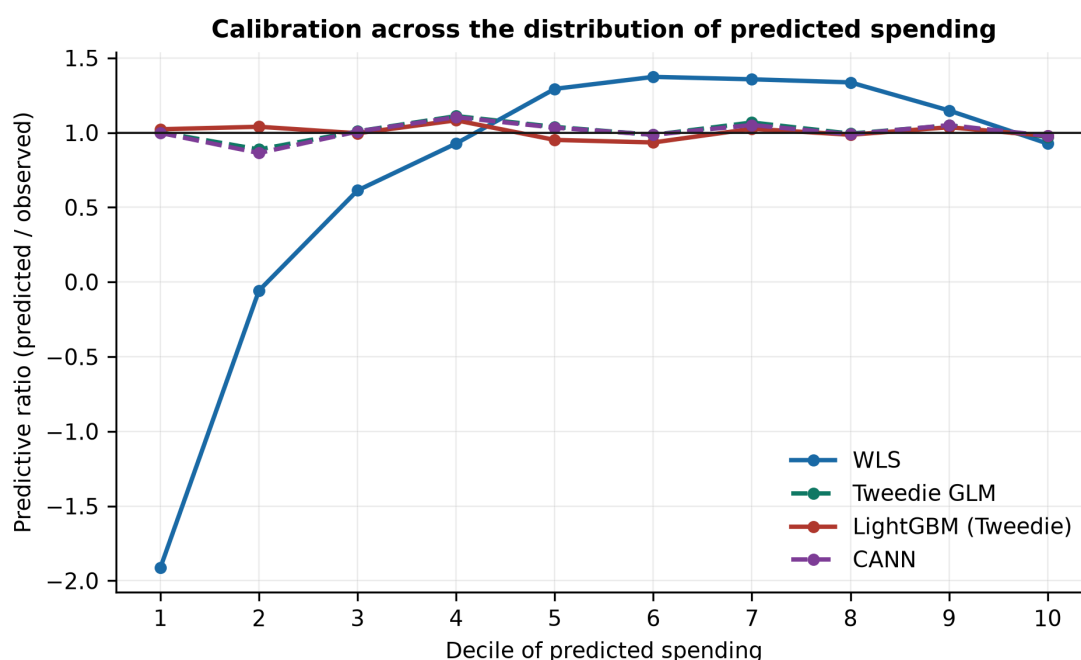


Figure 2. Predictive ratio by decile of predicted spending, four models.

Table 3 shows where the information is. On demographics alone every model reaches R^2 of 0.039. Adding conditions raises WLS to 0.103 and boosting to 0.102; adding prior use and spending raises WLS to 0.125 and boosting to 0.195. The boosting advantage therefore comes almost entirely from how it uses prior use, which payment formulas exclude by design: on F2, the set closest to a deployable formula, the flexible model has no advantage over the linear one. Adding social-risk features (F4) changes accuracy by less than 0.005 for any model.

6.2 Who is underpaid (RQ2)

Table 4 and Figure 3 report net compensation by group. One result dominates. People who need help with ADL or IADL, 2.7% of the population with a mean cost of \$29,881, are underpaid by every model: \$7,794 per person-year by WLS (interval \$4,584 to \$11,613), \$4,949 by the Tweedie GLM, \$5,670 by Tweedie LightGBM, and between \$4,770 and \$9,508 by the others. Flexible models do not remove the gap, because functional status is not in the primary features and no combination of diagnoses and use reproduces it.

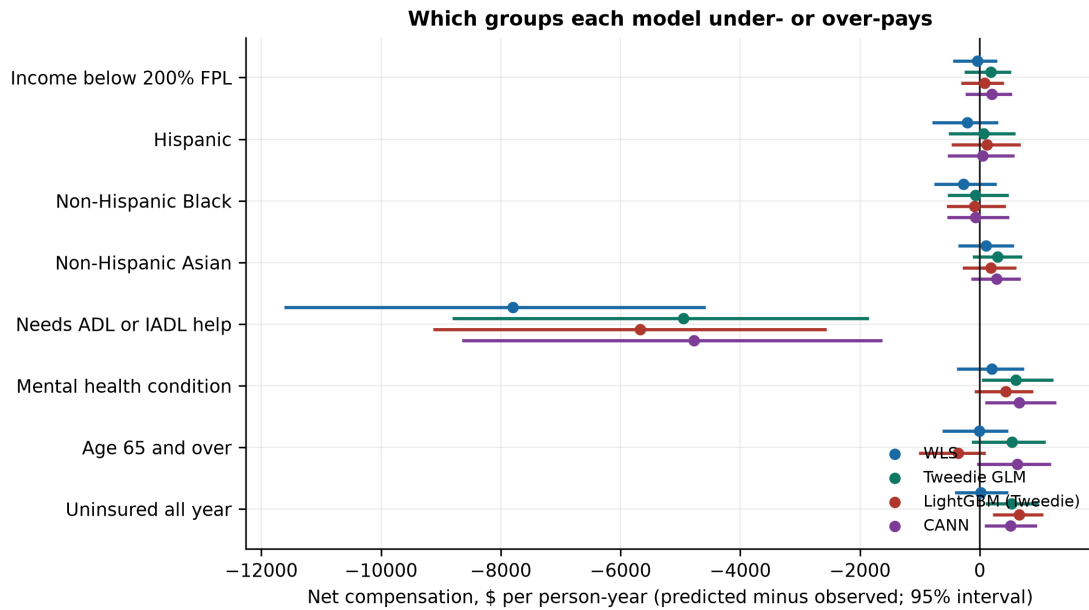


Figure 3. Net compensation by group and model, \$ per person-year, with 95% intervals.

The other groups are close to balance under most models, with intervals that include zero. WLS underpays Hispanic and non-Hispanic Black enrollees by \$207 and \$274 and the intervals include zero; the tree models pay these groups slightly above cost. People with a mental health condition are paid \$199 to \$793 above cost by most models. Enrollees aged 65 and over are underpaid by \$357 by Tweedie LightGBM and \$1,023 by squared-error LightGBM, with wide intervals.

6.3 The price of fairness (RQ2)

Table 5 and Figure 4 trace the frontier. For the linear formula, driving the ADL gap from \$7,887 to \$805 (its out-of-fold floor under the constraint) reduces R^2 from 0.124 to 0.096, a loss of 23%. For the stacked estimator, driving the gap from \$5,142 to \$210 reduces R^2 from 0.190 to 0.178, a loss of 6%. At every level of the gap the stacked frontier lies above the linear one. Decile recalibration of the boosted model, which uses no group information, leaves the gap at \$5,647.

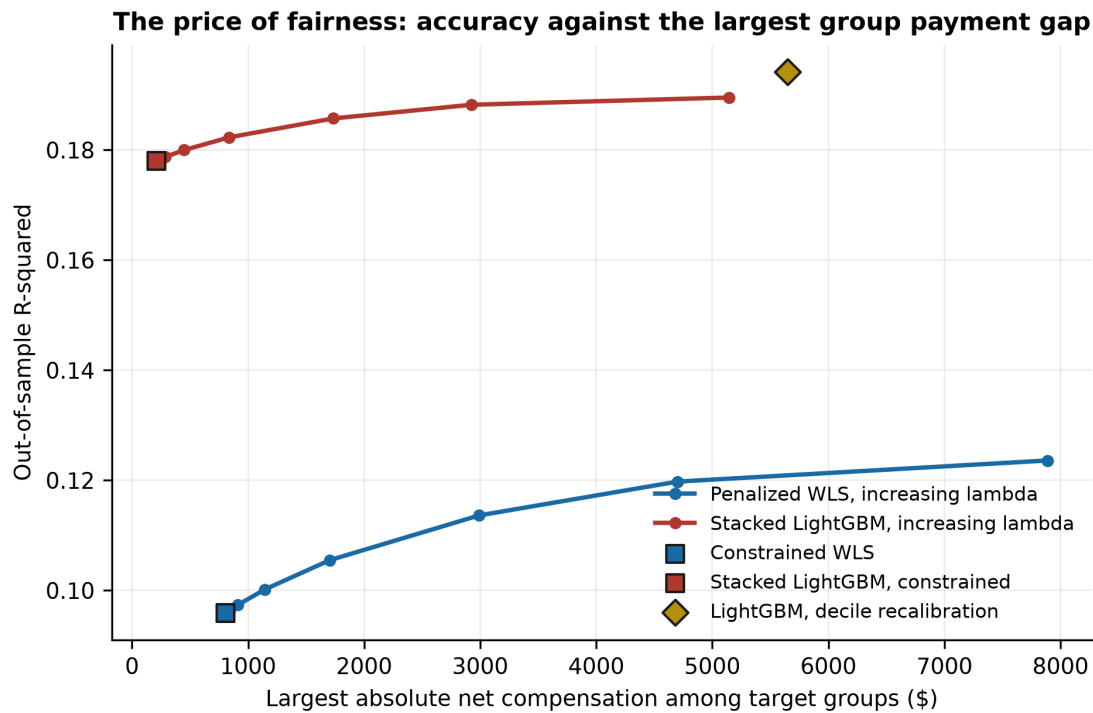


Figure 4. The accuracy-fairness frontier: out-of-sample R^2 against the largest absolute net compensation among target groups, for penalized and constrained linear and stacked estimators.

Closing the target gaps opens another. Under both constrained estimators, enrollees aged 65 and over move from balance to an overpayment of \$3,342 (linear) and \$2,156 (stacked), and non-Hispanic Black enrollees move to \$696 and \$439 below cost. The ADL group is largely elderly, and a constraint that raises payment for people who need help shifts payment towards the elderly generally. This spillover is invisible if only target groups are reported. We report all groups at every point of the frontier.

6.4 Drivers (RQ3)

Table 6 and Figure 5 list the largest contributors to the squared-error LightGBM. Year-1 spending is first by a wide margin (mean absolute SHAP \$3,970), followed by prescription fills (\$1,625), the condition count (\$1,015), office visits (\$557), inpatient discharges (\$386) and the body-system count (\$311). The first diagnosis category is diabetes without complication (\$172), then age 55 to 64. Importance ranks are stable across folds (Spearman 0.80 over all features, 0.77 over the top 30). SHAP contributions rise monotonically with year-1 spending, condition count and body-system count (rank correlations 0.88, 0.90 and 0.86; Table A4). For the CANN, permuting year-1 spending costs 0.108 of R^2 and no other feature costs more than 0.02.

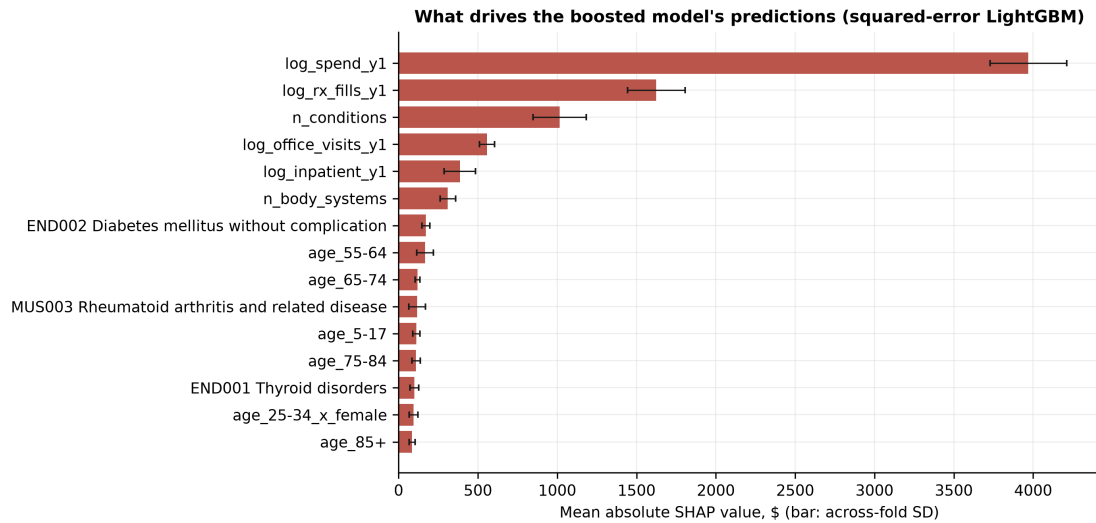


Figure 5. The fifteen largest drivers of the squared-error LightGBM predictions, mean absolute SHAP value in dollars with across-fold standard deviation.

6.5 Coding sensitivity (RQ4)

Table 7 and Figure 6 give the rise in predicted spending per added chronic code when 10% of people gain one. The linear formula rises most, \$1,292 per code; the two-part model \$1,307; the constrained linear formula \$1,121; the CANN \$953; the Tweedie GLM \$885; squared-error LightGBM \$755; and Tweedie LightGBM least, \$693. For codes drawn by prevalence rather than from the chronic pool (full grid in Table A3), the ordering changes: the tree models rise by \$612 to \$719 and the linear models by \$333 to \$443, because a linear formula assigns small coefficients to common low-cost categories while the trees respond to the change in the condition count.

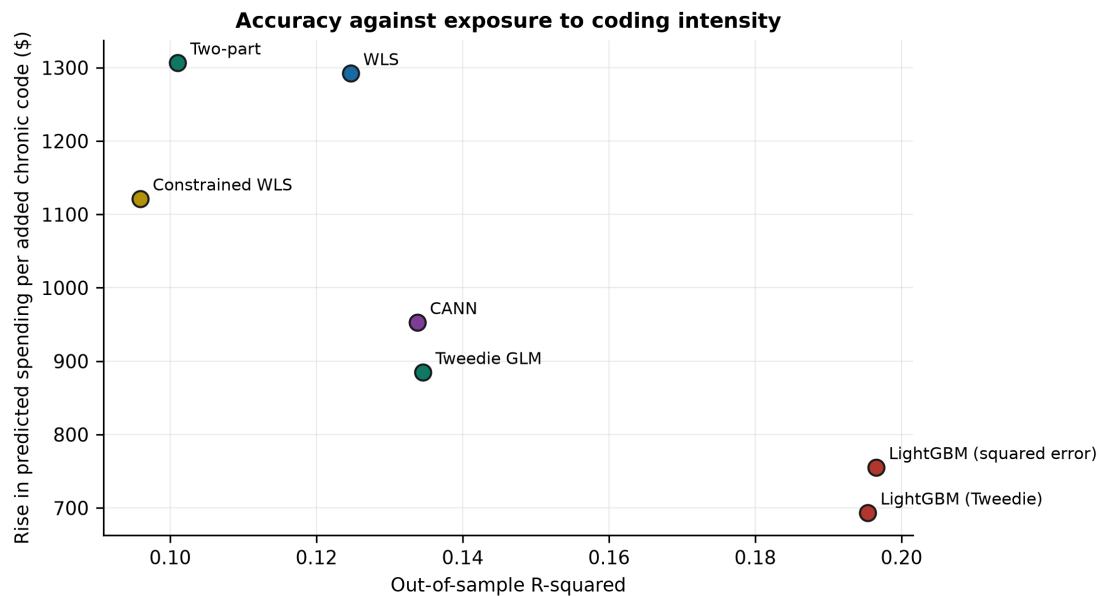


Figure 6. Accuracy against exposure to coding intensity: out-of-sample R^2 and the rise in predicted spending per added chronic code when 10% of people gain one.

The reason the linear formula is more exposed to chronic codes is structural. A linear model attaches a fixed increment to each flag regardless of what else is recorded. The tree models condition on prior use: an added diagnosis for someone whose visits and fills show no sign of it moves the prediction less. That is the same property that makes the trees more accurate.

6.6 Robustness

Table 8 reports the variants. No variant changes the ordering of the three models on R^2 , and the boosting advantage over WLS is between 0.06 and 0.09 in every one. Capping the outcome raises every R^2 (boosting to 0.282) and reduces the ADL gap by about \$1,300 for each model. Dropping prior spending (F2) collapses boosting's advantage and roughly doubles the ADL gap to \$10,670, since prior use was carrying part of the functional-status signal. Adding social-risk features (F4), which include the ADL and IADL items themselves, cuts the ADL gap to \$2,190 for boosting and \$1,462 for WLS: the direct route to paying for functional need is to observe it. Removing the pandemic outcome years, using unweighted training, and moving the condition-flag floor change R^2 by less than 0.02. The log-outcome linear model performs poorly on squared error ($R^2 -0.105$) and is not a candidate for payment. Within the 65-and-over subset, R^2 falls for every model (boosting 0.170, WLS 0.082) and the ADL gap persists.

7. Discussion

For payers and regulators. On this benchmark the choice between a linear formula and a flexible model is not a choice between accuracy and fairness. The flexible model is more accurate, is no worse on group compensation, and is less responsive to added chronic diagnoses. Its accuracy advantage depends on prior utilization, which payment formulas exclude to avoid rewarding use; a regulator who keeps that exclusion should expect the flexible model's advantage to vanish (Table 3) and its coding sensitivity to converge on the linear model's.

Two findings apply to any formula. First, the group that every model underpays is defined by functional status, which no diagnosis or use variable reproduces. The robustness results show two routes to paying for it: observe it directly (F4), or constrain payment to it, which costs accuracy and shifts money to the elderly. Second, fairness reporting must cover non-target groups. The constraint that fixed the ADL gap created an elderly overpayment of the same order, and a reader shown only the target groups would call the constrained formula fair.

Limitations. MEPS is a household survey with self-reported conditions verified by provider records; R^2 levels are not comparable with claims-based studies and comparisons are valid only within the benchmark. Three-digit ICD-10 codes make the condition categories coarser than payment-model categories. Sample sizes for small groups are limited; the ADL group has 1,488 persons and its intervals are wide. The coding simulation adds codes at random within a pool; real coding intensity is targeted at the codes that pay most, so the dollar figures are a lower bound on what targeted coding would do to the linear formula and the ordering across models is the more reliable result. The fairness targets were chosen by the analyst; other choices would move other groups. The neural models were tuned over a small grid and their results should be read as a floor on what a well-tuned network can do, not a ceiling.

What the toolkit is for. Any payer or regulator with person-level cost data can run the same four evaluations on their own formula and candidate replacements, with cluster-aware validation and group reporting built in. The package does not decide what fairness means; it measures what a formula does to each group and what changing that would cost.

8. Conclusion

A public benchmark shows that flexible models predict next-year health spending more accurately than the linear form of payment formulas, that the gain comes from prior utilization rather than from diagnoses, that every model underpays people who need functional help, that a stacked constrained estimator closes that gap at a small cost while shifting payment towards the elderly, and that the linear formula is the most responsive to added chronic codes. The code that produced every number is released as a reusable package.

Declarations

Data availability. MEPS public-use files are freely available from the Agency for Healthcare Research and Quality; the CCSR reference file from AHRQ's HCUP. Neither is redistributed.

Code availability. The `riskfair` package and the complete, seeded pipeline are at <https://github.com/tosinbabs/fair-ml-health-risk-adjustment>; `python/run_all.py` rebuilds every table and figure and `pytest tests` runs the checks. An interactive explorer of the results is at <https://fair-risk-adjustment.vercel.app>.

Competing interests. None declared.

Ethics. The analysis uses de-identified public survey data and did not require ethical approval.

AI-assistance disclosure. Generative AI (Claude, Anthropic) was used to assist with code development, code review and language editing. The author designed the study, specified all models and parameters, verified and interpreted all results, and takes full responsibility for the content. AI systems are not authors.

CRedit statement. Oluwatosin Dorcas Babalola: conceptualization, methodology, software, formal analysis, data curation, writing (original draft), writing (review and editing).

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Tables

Generated from `output/tables/*.csv` by `python/make_tables.py`. Spending is total expenditure from all payers in 2024 dollars. All estimates are weighted by the MEPS longitudinal weight; model metrics are out of fold from cross-validation with primary sampling units kept whole.

Table 1. Sample characteristics and year-2 spending.

Weighted estimates; standard errors Taylor-linearized for the stratified cluster design.

Characteristic	Estimate	SE	n
Persons	43,568.00		43,568
Weighted population, millions a year	324.38		43,568
Age in year 1, mean	38.78	0.20	43,568
Age 65 and over, %	16.24	0.27	43,568
Female, %	50.92	0.24	43,568
Hispanic, %	19.06	0.60	43,568
Non-Hispanic White, %	58.57	0.64	43,568
Non-Hispanic Black, %	12.44	0.42	43,568
Non-Hispanic Asian, %	6.16	0.30	43,568
Non-Hispanic other or multiple, %	3.76	0.19	43,568
Below 100% FPL, %	12.29	0.34	43,568
100-124% FPL, %	4.06	0.20	43,568
125-199% FPL, %	12.27	0.34	43,568
200-399% FPL, %	28.91	0.48	43,568
400% FPL or more, %	42.48	0.58	43,568
Any private coverage, %	65.95	0.50	43,568
Public coverage only, %	27.35	0.44	43,568
Uninsured all year, %	6.70	0.24	43,568
Fair or poor perceived health, %	9.64	0.21	43,396
Needs help with ADL or IADL, %	2.72	0.10	43,568

Characteristic	Estimate	SE	n
Any year-1 condition (CCSR category), %	69.95	0.39	43,568
Year-1 CCSR categories, mean	2.79	0.03	43,568
Any mental health condition, %	15.80	0.28	43,568
Year-1 spending, mean \$	7,285.40	157.46	43,568
Year-2 spending, mean \$	7,538.02	143.02	43,568
Year-2 spending of zero, %	14.97	0.32	43,568
Year-2 spending, median \$	1,486.14		43,568
Year-2 spending, 90th percentile \$	18,256.33		43,568
Year-2 spending, 99th percentile \$	91,441.62		43,568
Share of year-2 spending by the top 5% of spenders, %	49.97		43,568

Table 2. Out-of-sample accuracy by model, primary feature set (F3).

Five survey folds, three repeats. Intervals from a bootstrap over PSUs within strata applied to repeat-averaged out-of-fold predictions. CPM is Cumming's prediction measure. Top-10% capture is the weighted share of the true top decile of spenders placed in the model's top decile.

Model	R ²	95% low	95% high	CPM	MAE	PR bottom decile	PR top decile	Top-10% capture
WLS	0.125	0.101	0.158	0.151	\$7,978	-1.91	0.93	44%
Tweedie GLM	0.135	0.104	0.180	0.247	\$7,077	1.00	0.97	47%
Two-part	0.101	0.077	0.134	0.217	\$7,359	0.98	0.99	44%
Elastic net	0.117	0.092	0.152	0.197	\$7,540	-0.86	0.93	42%
LightGBM (Tweedie)	0.195	0.161	0.245	0.279	\$6,770	1.02	0.98	48%
LightGBM (squared error)	0.197	0.165	0.240	0.268	\$6,881	1.58	0.97	48%
Random forest	0.197	0.167	0.241	0.270	\$6,856	0.75	0.97	48%
Neural network	0.141	0.116	0.178	0.249	\$7,053	1.00	0.97	44%
CANN	0.134	0.103	0.179	0.247	\$7,074	1.00	0.98	47%

Table 3. Out-of-sample R² by feature set.

F1 demographics; F2 adds year-1 CCSR condition flags; F3 adds year-1 use and spending; F4 adds income, insurance, perceived health and functional help.

Model	F1	F2	F3	F4
LightGBM (Tweedie)	0.039	0.102	0.195	0.195
Tweedie GLM	0.039	0.013	0.135	0.132
WLS	0.039	0.103	0.125	0.129

Table 4. Net compensation by group: predicted minus observed spending, \$ per person-year (95% interval).

Negative values mean the model pays less for the group than the group costs. Race and ethnicity are used for evaluation only and never as model inputs.

Group	WLS	Tweedie GLM	LightGBM (Tweedie)	CANN
Age 65 and over	-\$8 (-\$624, \$473)	\$538 (-\$133, \$1,097)	-\$357 (-\$1,014, \$97)	\$627 (-\$48, \$1,189)
Hispanic	-\$207 (-\$793, \$308)	\$66 (-\$522, \$597)	\$124 (-\$472, \$679)	\$51 (-\$536, \$582)
Income below 200% FPL	-\$37 (-\$451, \$288)	\$187 (-\$256, \$525)	\$83 (-\$311, \$404)	\$201 (-\$241, \$538)
Mental health condition	\$199 (-\$381, \$743)	\$605 (\$32, \$1,226)	\$433 (-\$91, \$895)	\$658 (\$87, \$1,275)
Needs ADL or IADL help	-\$7,794 (-\$11,613, -\$4,584)	-\$4,949 (-\$8,810, -\$1,852)	-\$5,670 (-\$9,127, -\$2,561)	-\$4,770 (-\$8,648, -\$1,630)
Non-Hispanic Asian	\$105 (-\$358, \$573)	\$300 (-\$123, \$706)	\$189 (-\$284, \$613)	\$282 (-\$143, \$686)
Non-Hispanic Black	-\$274 (-\$760, \$280)	-\$68 (-\$534, \$486)	-\$89 (-\$550, \$437)	-\$72 (-\$543, \$489)
Uninsured all year	\$17 (-\$413, \$475)	\$533 (\$97, \$979)	\$657 (\$215, \$1,061)	\$513 (\$79, \$959)

Table 5. The accuracy-fairness frontier.

Penalized and constrained estimators target the three groups marked in the text; the other groups show spillover. Stacked LightGBM enters a cross-fitted LightGBM score with the features into the fair regression. One repeat of the survey folds.

Method	λ	R ²	CPM	Max	NC	target	Income below 200% FPL	Hispanic	Non-Hispanic Black	Non-Hispanic Asian	Needs ADL or IADL help	Mental health condition
Penalized WLS	0	0.124	0.149	\$7,887	-\$43	-\$215	-\$293	\$97	-\$7,887	\$174	\$15	\$12
Penalized WLS	0.1	0.120	0.118	\$4,699	\$585	-\$29	\$118	\$290	-\$4,699	\$1,054	\$1,653	\$44
Penalized WLS	0.3	0.114	0.102	\$2,991	\$701	-\$158	\$140	\$216	-\$2,991	\$874	\$2,516	-\$166
Penalized WLS	1	0.105	0.084	\$1,703	\$517	-\$554	-\$118	-\$85	-\$1,703	\$428	\$3,089	-\$639
Penalized WLS	3	0.100	0.071	\$1,145	\$268	-\$916	-\$410	-\$376	-\$1,145	\$180	\$3,272	-\$1,051
Penalized WLS	10	0.097	0.063	\$912	\$106	-\$1,132	-\$594	-\$552	-\$912	\$74	\$3,325	-\$1,293
Penalized WLS	30	0.096	0.060	\$841	\$47	-\$1,209	-\$660	-\$615	-\$841	\$42	\$3,337	-\$1,379
Penalized WLS	100	0.096	0.059	\$816	\$25	-\$1,238	-\$685	-\$638	-\$816	\$30	\$3,341	-\$1,411

Method	λ	R^2	CPM	Max	NC	target	Income below 200% FPL	Hispanic	Non- Hispanic Black	Non- Hispanic Asian	Needs ADL or IADL help	Mental health condition
Constrained WLS	constrained	0.096	0.058	\$805	\$15	-\$1,250	-\$696	-\$649	-\$805	\$25	\$3,342	-\$1,425
Stacked LightGBM	0	0.190	0.258	\$5,142	\$71	\$60	-\$91	\$201	-\$5,142	\$165	-\$18	\$441
Stacked LightGBM	0.1	0.188	0.244	\$2,925	\$486	\$186	\$182	\$329	-\$2,925	\$769	\$1,080	\$467
Stacked LightGBM	0.3	0.186	0.240	\$1,736	\$549	\$91	\$184	\$270	-\$1,736	\$654	\$1,648	\$320
Stacked LightGBM	1	0.182	0.236	\$838	\$400	-\$193	-\$13	\$50	-\$838	\$361	\$2,013	-\$17
Stacked LightGBM	3	0.180	0.231	\$448	\$213	-\$454	-\$229	-\$162	-\$448	\$198	\$2,121	-\$312
Stacked LightGBM	10	0.179	0.228	\$286	\$94	-\$609	-\$364	-\$290	-\$286	\$130	\$2,148	-\$486
Stacked LightGBM	30	0.178	0.227	\$236	\$50	-\$664	-\$413	-\$335	-\$236	\$109	\$2,154	-\$547
Stacked LightGBM	100	0.178	0.227	\$218	\$34	-\$685	-\$431	-\$353	-\$218	\$101	\$2,155	-\$570
Stacked LightGBM, constrained	constrained	0.178	0.226	\$210	\$27	-\$694	-\$439	-\$360	-\$210	\$98	\$2,156	-\$580
LightGBM, decile recalibration	0	0.194	0.275	\$5,647	\$110	\$162	-\$42	\$191	-\$5,647	\$555	-\$387	\$666

Table 6. The 20 largest drivers of the squared-error LightGBM predictions.

Weighted mean absolute SHAP value on held-out folds. Rank stability across folds: mean Spearman correlation 0.80 over all features and 0.77 over the top 30. The last column is the loss in held-out R^2 when the feature is permuted in the CANN.

Rank	Feature	Mean	SHAP	
1	log_spend_y1	\$3,970	\$242	0.1083
2	log_rx_fills_y1	\$1,625	\$181	0.0022
3	n_conditions	\$1,015	\$168	0.0205
4	log_office_visits_y1	\$557	\$47	-0.0019
5	log_inpatient_y1	\$386	\$98	0.0168
6	n_body_systems	\$311	\$48	-0.0083
7	END002 Diabetes mellitus without complication	\$172	\$26	-0.0009
8	age_55-64	\$167	\$53	0.0143
9	age_65-74	\$119	\$15	0.0025
10	MUS003 Rheumatoid arthritis and related disease	\$116	\$52	-0.0131
11	age_5-17	\$112	\$22	-0.0082

Rank	Feature	Mean	SHAP	
12	age_75-84	\$110	\$27	-0.0030
13	END001 Thyroid disorders	\$100	\$27	0.0035
14	age_25-34_x_female	\$95	\$27	0.0096
15	age_85+	\$84	\$19	0.0057
16	log_er_visits_y1	\$74	\$31	0.0022
17	region_3.0	\$73	\$28	0.0028
18	END010 Disorders of lipid metabolism	\$72	\$30	0.0110
19	MUS006 Osteoarthritis	\$69	\$25	-0.0000
20	region_2.0	\$65	\$24	-0.0013

Table 7. Coding sensitivity: rise in predicted spending per added condition code, 10% of people affected.

Codes the person did not have are switched on; use and spending are unchanged. The last column is the rise in total predicted spending when 10% of people gain one chronic code.

Model	chronic, 1 code	chronic, 2 codes	prevalence, 1 code	prevalence, 2 codes	Total rise, 1 chronic code
CANN	\$953	\$952	\$482	\$398	1.25%
Constrained WLS	\$1,121	\$1,228	\$333	\$383	1.54%
LightGBM (Tweedie)	\$693	\$755	\$612	\$596	0.92%
LightGBM (squared error)	\$755	\$781	\$719	\$717	1.00%
Tweedie GLM	\$885	\$889	\$443	\$358	1.16%
Two-part	\$1,307	\$1,402	\$827	\$842	1.73%
WLS	\$1,292	\$1,368	\$395	\$393	1.71%

Table 8. Robustness: each row changes one decision.

One repeat of the survey folds. Max |NC| is over the three target groups.

Variant	Model	n	R ²	CPM	PR top decile	Max
baseline	WLS	43,568	0.124	0.149	0.94	\$7,887
baseline	Tweedie GLM	43,568	0.125	0.243	1.00	\$5,011
baseline	LightGBM (Tweedie)	43,568	0.195	0.277	0.99	\$5,864
outcome capped	WLS	43,568	0.189	0.173	0.93	\$6,699
outcome capped	Tweedie GLM	43,568	0.194	0.250	1.00	\$4,034
outcome capped	LightGBM (Tweedie)	43,568	0.282	0.284	1.00	\$4,558
log outcome	WLS, log outcome	43,568	-0.105	0.195	1.64	\$6,407

Variant	Model	n	R ²	CPM	PR top decile	Max
unweighted training	WLS	43,568	0.127	0.149	0.91	\$7,806
unweighted training	Tweedie GLM	43,568	0.144	0.257	0.97	\$5,223
unweighted training	LightGBM (Tweedie)	43,568	0.196	0.283	0.99	\$5,356
CCSR floor 0.2%	WLS	43,568	0.125	0.155	1.00	\$7,369
CCSR floor 0.2%	Tweedie GLM	43,568	0.125	0.247	1.02	\$4,531
CCSR floor 0.2%	LightGBM (Tweedie)	43,568	0.197	0.280	0.99	\$5,685
CCSR floor 1%	WLS	43,568	0.124	0.150	0.90	\$8,668
CCSR floor 1%	Tweedie GLM	43,568	0.157	0.251	0.97	\$5,868
CCSR floor 1%	LightGBM (Tweedie)	43,568	0.195	0.277	0.99	\$5,824
no prior spending (F2)	WLS	43,568	0.101	0.189	1.06	\$8,908
no prior spending (F2)	Tweedie GLM	43,568	0.013	0.176	1.24	\$5,223
no prior spending (F2)	LightGBM (Tweedie)	43,568	0.101	0.199	1.01	\$10,670
social-risk features (F4)	WLS	43,568	0.127	0.152	0.96	\$1,462
social-risk features (F4)	Tweedie GLM	43,568	0.122	0.245	1.03	\$1,236
social-risk features (F4)	LightGBM (Tweedie)	43,568	0.194	0.280	0.99	\$2,190
decedents included	WLS	43,963	0.128	0.151	0.94	\$6,773
decedents included	Tweedie GLM	43,963	0.129	0.243	1.01	\$3,896
decedents included	LightGBM (Tweedie)	43,963	0.193	0.277	0.99	\$4,529
pandemic outcomes out	WLS	28,208	0.124	0.146	0.97	\$9,652
pandemic outcomes out	Tweedie GLM	28,208	0.108	0.249	1.03	\$6,659
pandemic outcomes out	LightGBM (Tweedie)	28,208	0.211	0.287	0.99	\$7,998
age 65 and over	WLS	9,173	0.082	0.080	1.08	\$5,587
age 65 and over	Tweedie GLM	9,173	0.109	0.164	1.07	\$3,547

Variant	Model	n	R ²	CPM	PR top decile	Max
age 65 and over	LightGBM (Tweedie)	9,173	0.170	0.197	0.93	\$4,944
under 65, private	WLS	21,962	0.091	0.095	1.02	\$7,994
under 65, private	Tweedie GLM	21,962	0.091	0.178	1.07	\$3,941
under 65, private	LightGBM (Tweedie)	21,962	0.160	0.211	0.97	\$6,579

Appendix tables

Table A1. Sample flow by MEPS panel.

Panel	Years	In file	Both years, weight > 0	Died in year 2	Analysis sample	Any year-1 condition
23	2018-2019	14,067	13,766	117	13,648	9,679
24	2019-2020	9,797	9,569	107	9,461	6,825
25	2020-2021	6,078	5,942	43	5,899	4,244
26	2021-2022	6,741	6,579	67	6,512	4,931
27	2022-2023	8,292	8,109	61	8,048	6,004

Table A2. Predictive ratio by decile of predicted spending.

Decile	CANN	Elastic net	LightGBM (Tweedie)	LightGBM (squared error)	Neural network	Random forest	Tweedie GLM	Two-part	WLS
1	1.00	-0.86	1.02	1.58	1.00	0.75	1.00	0.98	-1.91
2	0.86	0.18	1.04	1.54	1.02	1.02	0.89	1.25	-0.06
3	1.01	0.68	1.00	1.10	0.92	1.06	1.01	1.41	0.61
4	1.10	0.93	1.08	1.03	1.01	1.02	1.11	1.42	0.93
5	1.03	1.24	0.95	0.99	0.92	1.06	1.04	1.37	1.29
6	0.99	1.44	0.93	1.01	0.97	0.94	0.98	1.07	1.37
7	1.05	1.27	1.03	0.96	0.97	1.06	1.07	0.97	1.36
8	0.99	1.17	0.99	0.95	0.96	1.03	0.99	0.88	1.34
9	1.05	0.96	1.04	1.01	1.03	1.03	1.05	0.89	1.15
10	0.98	0.93	0.98	0.97	0.97	0.97	0.97	0.99	0.93

Table A3. Coding sensitivity, full grid.

Model	Pool	Share affected	Codes added	Total rise	\$ per code
WLS	prevalence	5%	1	0.26%	\$394
WLS	prevalence	5%	2	0.53%	\$408
WLS	prevalence	10%	1	0.52%	\$395
WLS	prevalence	10%	2	1.05%	\$393
WLS	prevalence	25%	1	1.16%	\$351
WLS	prevalence	25%	2	2.50%	\$376

Model	Pool	Share affected	Codes added	Total rise	\$ per code
WLS	chronic	5%	1	0.80%	\$1,216
WLS	chronic	5%	2	1.76%	\$1,351
WLS	chronic	10%	1	1.71%	\$1,292
WLS	chronic	10%	2	3.64%	\$1,368
WLS	chronic	25%	1	4.25%	\$1,283
WLS	chronic	25%	2	8.99%	\$1,352
Tweedie GLM	prevalence	5%	1	0.24%	\$375
Tweedie GLM	prevalence	5%	2	0.58%	\$449
Tweedie GLM	prevalence	10%	1	0.58%	\$443
Tweedie GLM	prevalence	10%	2	0.95%	\$358
Tweedie GLM	prevalence	25%	1	1.19%	\$361
Tweedie GLM	prevalence	25%	2	2.41%	\$364
Tweedie GLM	chronic	5%	1	0.56%	\$854
Tweedie GLM	chronic	5%	2	1.22%	\$942
Tweedie GLM	chronic	10%	1	1.16%	\$885
Tweedie GLM	chronic	10%	2	2.36%	\$889
Tweedie GLM	chronic	25%	1	2.83%	\$861
Tweedie GLM	chronic	25%	2	5.94%	\$898
Two-part	prevalence	5%	1	0.53%	\$806
Two-part	prevalence	5%	2	1.21%	\$929
Two-part	prevalence	10%	1	1.09%	\$827
Two-part	prevalence	10%	2	2.24%	\$842
Two-part	prevalence	25%	1	2.66%	\$803
Two-part	prevalence	25%	2	5.62%	\$846
Two-part	chronic	5%	1	0.85%	\$1,299
Two-part	chronic	5%	2	1.90%	\$1,463
Two-part	chronic	10%	1	1.73%	\$1,307
Two-part	chronic	10%	2	3.73%	\$1,402
Two-part	chronic	25%	1	4.28%	\$1,294
Two-part	chronic	25%	2	9.26%	\$1,393
LightGBM (Tweedie)	prevalence	5%	1	0.40%	\$602
LightGBM (Tweedie)	prevalence	5%	2	0.80%	\$612
LightGBM (Tweedie)	prevalence	10%	1	0.81%	\$612
LightGBM (Tweedie)	prevalence	10%	2	1.59%	\$596
LightGBM (Tweedie)	prevalence	25%	1	1.96%	\$590
LightGBM (Tweedie)	prevalence	25%	2	4.09%	\$612

Model	Pool	Share affected	Codes added	Total rise	\$ per code
LightGBM (Tweedie)	chronic	5%	1	0.47%	\$718
LightGBM (Tweedie)	chronic	5%	2	0.98%	\$750
LightGBM (Tweedie)	chronic	10%	1	0.92%	\$693
LightGBM (Tweedie)	chronic	10%	2	2.02%	\$755
LightGBM (Tweedie)	chronic	25%	1	2.37%	\$713
LightGBM (Tweedie)	chronic	25%	2	5.00%	\$749
LightGBM (squared error)	prevalence	5%	1	0.47%	\$708
LightGBM (squared error)	prevalence	5%	2	0.96%	\$739
LightGBM (squared error)	prevalence	10%	1	0.95%	\$719
LightGBM (squared error)	prevalence	10%	2	1.92%	\$717
LightGBM (squared error)	prevalence	25%	1	2.26%	\$679
LightGBM (squared error)	prevalence	25%	2	4.87%	\$729
LightGBM (squared error)	chronic	5%	1	0.47%	\$721
LightGBM (squared error)	chronic	5%	2	1.02%	\$784
LightGBM (squared error)	chronic	10%	1	1.00%	\$755
LightGBM (squared error)	chronic	10%	2	2.09%	\$781
LightGBM (squared error)	chronic	25%	1	2.46%	\$742
LightGBM (squared error)	chronic	25%	2	5.29%	\$792
CANN	prevalence	5%	1	0.27%	\$408
CANN	prevalence	5%	2	0.65%	\$501
CANN	prevalence	10%	1	0.63%	\$482
CANN	prevalence	10%	2	1.05%	\$398
CANN	prevalence	25%	1	1.30%	\$396
CANN	prevalence	25%	2	2.65%	\$400
CANN	chronic	5%	1	0.59%	\$901
CANN	chronic	5%	2	1.31%	\$1,013
CANN	chronic	10%	1	1.25%	\$953
CANN	chronic	10%	2	2.52%	\$952

Model	Pool	Share affected	Codes added	Total rise	\$ per code
CANN	chronic	25%	1	3.03%	\$920
CANN	chronic	25%	2	6.36%	\$961
Constrained WLS	prevalence	5%	1	0.27%	\$400
Constrained WLS	prevalence	5%	2	0.51%	\$376
Constrained WLS	prevalence	10%	1	0.46%	\$333
Constrained WLS	prevalence	10%	2	1.06%	\$383
Constrained WLS	prevalence	25%	1	0.96%	\$280
Constrained WLS	prevalence	25%	2	2.38%	\$344
Constrained WLS	chronic	5%	1	0.66%	\$970
Constrained WLS	chronic	5%	2	1.64%	\$1,216
Constrained WLS	chronic	10%	1	1.54%	\$1,121
Constrained WLS	chronic	10%	2	3.40%	\$1,228
Constrained WLS	chronic	25%	1	3.82%	\$1,112
Constrained WLS	chronic	25%	2	8.26%	\$1,194

Table A4. Actuarial sense checks on the SHAP values.

Spearman correlation between a feature's value and its SHAP contribution across held-out persons, by fold. Positive means predicted spending rises with the feature.

Feature	Mean	Min across folds	Max across folds
log_spend_y1	0.875	0.859	0.903
n_body_systems	0.863	0.842	0.889
n_conditions	0.896	0.873	0.915

Table A5. Hyperparameters chosen by inner validation, across the 15 outer folds.

model	parameter	chosen (count of folds)
tweedie	alpha	1.0 (6); 10.0 (4); 0.01 (3); 0.1 (2)
twopart	C	0.1 (15)
twopart	alpha	1.0 (11); 0.1 (3); 10.0 (1)
enet	alpha	1.0 (15)
enet	l1_ratio	0.8 (15)
gbm	num_leaves	63 (8); 31 (4); 15 (3)
gbm	min_child_samples	100 (11); 50 (4)

model	parameter	chosen (count of folds)
gbm	n_estimators	152 (2); 165 (1); 119 (1); 145 (1); 121 (1); 111 (1);
		136 (1); 170 (1); 212 (1); 215 (1); 132 (1); 113 (1); 102 (1); 104 (1)
gbm_mse	num_leaves	31 (7); 63 (6); 15 (2)
gbm_mse	min_child_samples	100 (8); 50 (7)
gbm_mse	n_estimators	88 (1); 124 (1); 98 (1); 162 (1); 62 (1); 73 (1); 76
		(1); 354 (1); 123 (1); 127 (1); 126 (1); 484 (1); 187 (1); 83 (1); 68 (1)
mlp	lr	0.003 (7); 0.01 (6); 0.001 (2)
mlp	hidden	(64, 32) (7); (32,) (6); (128, 64) (2)
mlp	epochs	32 (4); 29 (4); 28 (2); 33 (2); 38 (1); 30 (1); 35 (1)
cann	lr	0.001 (11); 0.01 (2); 0.003 (2)
cann	hidden	(64, 32) (11); (32,) (2); (128, 64) (2)
cann	epochs	25 (10); 26 (5)

Table A6. Year-1 CCSR categories entering the feature sets.

Categories held by at least 0.5% of the sample in year 1. MEPS body-system placeholder codes (XXX000) are excluded.

CCSR	Description	Prevalence
CIR007	Essential hypertension	22.13%
END010	Disorders of lipid metabolism	16.78%
MUS010	Musculoskeletal pain, not low back pain	9.59%
END005	Diabetes mellitus, Type 2	9.53%
END002	Diabetes mellitus without complication	9.53%
MBD005	Anxiety and fear-related disorders	8.03%
MBD002	Depressive disorders	7.06%
END001	Thyroid disorders	7.01%
MUS006	Osteoarthritis	6.68%
RSP009	Asthma	6.32%
DIG004	Esophageal disorders	5.89%
NVS016	Sleep wake disorders	5.41%
SKN007	Other specified and unspecified skin disorders	4.94%
FAC016	Exposure, encounters, screening or contact with infectious disease	4.78%
MUS011	Spondylopathies/spondyloarthropathy (including infective)	4.48%
RSP006	Other specified upper respiratory infections	4.24%
INJ031	Allergic reactions	3.92%
SYM016	Other general signs and symptoms	3.76%
RSP007	Other specified and unspecified upper respiratory disease	3.67%
MBD014	Neurodevelopmental disorders	3.11%
INJ067	Allergic reactions, subsequent encounter	3.11%

CCSR	Description	Prevalence
INJ064	Other unspecified injuries, subsequent encounter	2.95%
CIR011	Coronary atherosclerosis and other heart disease	2.93%
MUS025	Other specified connective tissue disease	2.93%
INF012	Coronavirus disease – 2019 (COVID-19)	2.90%
GEN004	Urinary tract infections	2.69%
NVS010	Headache; including migraine	2.68%
SYM013	Respiratory signs and symptoms	2.67%
RSP003	Influenza	2.56%
INF008	Viral infection	2.53%
SKN002	Other specified inflammatory condition of skin	2.44%
EYE002	Cataract and other lens disorders	2.31%
SYM006	Abdominal pain and other digestive/abdomen signs and symptoms	2.30%
RSP001	Sinusitis	2.29%
MUS038	Low back pain	2.25%
CIR017	Cardiac dysrhythmias	2.24%
EAR001	Otitis media	2.07%
DIG025	Other specified and unspecified gastrointestinal disorders	2.07%
END007	Nutritional deficiencies	2.04%
FAC014	Medical examination/evaluation	2.02%
INF003	Bacterial infections	2.01%
INJ061	Sprains and strains, subsequent encounter	2.01%
SYM010	Nervous system signs and symptoms	2.01%
FAC003	Encounter for observation and examination for conditions ruled out (excludes infectious disease, neoplasm, mental disorders)	1.95%
SYM014	Skin/Subcutaneous signs and symptoms	1.94%
DIG002	Disorders of teeth and gingiva	1.93%
FAC012	Other specified encounters and counseling	1.91%
SYM017	Abnormal findings without diagnosis	1.89%
MBD007	Trauma- and stressor-related disorders	1.76%
FAC009	Implant, device or graft related encounter	1.75%
EYE003	Glaucoma	1.73%
NEO028	Skin cancers - all other types	1.70%
SYM012	Circulatory signs and symptoms	1.69%
FAC013	Contraceptive and procreative management	1.68%
EYE009	Refractive error	1.67%
RSP008	Chronic obstructive pulmonary disease and bronchiectasis	1.66%

CCSR	Description	Prevalence
MUS003	Rheumatoid arthritis and related disease	1.62%
CIR009	Acute myocardial infarction	1.54%
MUS007	Other specified joint disorders	1.50%
EYE005	Retinal and vitreous conditions	1.42%
FAC025	Other specified status	1.38%
INF004	Fungal infections	1.38%
RSP005	Acute bronchitis	1.34%
PRG029	Uncomplicated pregnancy, delivery or puerperium	1.30%
EAR006	Other specified and unspecified disorders of the ear	1.29%
DIG008	Other specified and unspecified disorders of stomach and duodenum	1.29%
END011	Fluid and electrolyte disorders	1.29%
NVS012	Transient cerebral ischemia	1.28%
MUS033	Gout	1.27%
NVS019	Nervous system pain and pain syndromes	1.26%
NVS017	Nerve and nerve root disorders	1.22%
SKN001	Skin and subcutaneous tissue infections	1.21%
NEO073	Benign neoplasms	1.21%
MUS009	Tendon and synovial disorders	1.18%
FAC008	Neoplasm-related encounters	1.18%
EAR004	Hearing loss	1.15%
EYE010	Blindness and vision defects	1.12%
SYM015	General sensation/perception signs and symptoms	1.11%
EYE008	Oculofacial plastics and orbital conditions	1.09%
GEN012	Hyperplasia of prostate	1.07%
SYM004	Nausea and vomiting	1.07%
RSP002	Pneumonia (except that caused by tuberculosis)	1.04%
GEN006	Other specified and unspecified diseases of kidney and ureters	0.98%
NVS015	Polyneuropathies	0.98%
EYE012	Other specified eye disorders	0.96%
END015	Other specified and unspecified endocrine disorders	0.93%
CIR030	Aortic and peripheral arterial embolism or thrombosis	0.93%
MUS028	Other specified bone disease and musculoskeletal deformities	0.92%
MBD003	Bipolar and related disorders	0.92%
NVS009	Epilepsy; convulsions	0.92%
GEN023	Menopausal disorders	0.91%

CCSR	Description	Prevalence
NEO030	Breast cancer - all other types	0.90%
GEN005	Calculus of urinary tract	0.85%
EYE001	Cornea and external disease	0.85%
NEO072	Neoplasms of unspecified nature or uncertain behavior	0.82%
INJ042	Fracture of lower limb (except hip), subsequent encounter	0.81%
MUS026	Muscle disorders	0.80%
INJ041	Fracture of the upper limb, subsequent encounter	0.80%
CIR012	Nonspecific chest pain	0.79%
SYM011	Genitourinary signs and symptoms	0.77%
SYM002	Fever	0.76%
MBD013	Miscellaneous mental and behavioral disorders/conditions	0.76%
CIR015	Other and ill-defined heart disease	0.76%
DIG010	Abdominal hernia	0.76%
FAC010	Other aftercare encounter	0.76%
RSP016	Other specified and unspecified lower respiratory disease	0.75%
GEN025	Other specified female genital disorders	0.73%
GEN016	Other specified male genital disorders	0.72%
NVS006	Other nervous system disorders (often hereditary or degenerative)	0.72%
GEN017	Nonmalignant breast conditions	0.68%
MUS013	Osteoporosis	0.67%
GEN008	Urinary incontinence	0.66%
INJ027	Other unspecified injury	0.66%
BLD003	Aplastic anemia	0.65%
DIG001	Intestinal infection	0.64%
NEO039	Male reproductive system cancers - prostate	0.63%
GEN007	Other specified and unspecified diseases of bladder and urethra	0.62%
INF009	Parasitic, other specified and unspecified infections	0.59%
NEO025	Skin cancers - melanoma	0.58%
DEN001	Any dental condition including traumatic injury	0.57%
INJ049	Open wounds to limbs, subsequent encounter	0.56%
DEN002	Nontraumatic dental conditions	0.53%
SKN005	Contact dermatitis	0.53%
DIG019	Other specified and unspecified liver disease	0.51%